GROUP 22

FINAL PROJECT TITLE: Convenience Store Market Basket Pattern Explorer
Scenario: A convenience store wants to analyze customer transaction records to discover products that are frequently purchased together. The system uses association-rule mining to identify common product combinations and purchasing patterns.

Dataset: 50 transactions, 15 products
Primary algorithm: Apriori
Comparison method: Frequent itemsets by support count
Development tools: KNIME, Python, Streamlit
KNIME — Analyzes the transactions visually through connected nodes.
Python — Recreates the same analysis using code.
Streamlit — Provides a simple website where users can upload transactions, adjust settings, and see the results.

ROLES BELOW:

Member	Primary responsibility	Secondary responsibility
Sison	Python + Streamlit	Project leadership and integration
Sablan	KNIME + Algorithm Evaluation	Results comparison
Satimbre	Dataset + Documentation	Testing and KNIME assistance

Dataset: Synthetic convenience-store purchases

Transactions: 50

Transaction IDs: T001–T050

Unique products: 15

Typical items per basket: 2–5

DOCUMENTATION BELOW:

PRODUCTS	
Product no. #	Product name
1	Soft Drink
2	Bread
3	Chips
4	Instant Noodles
5	Coffee
6	Milk
7	Biscuits
8	Eggs
9	Corned Beef
10	Sardines
11	Bottled Water
12	Iced Tea
13	Chocolate
14	Peanut Butter
15	Hotdog

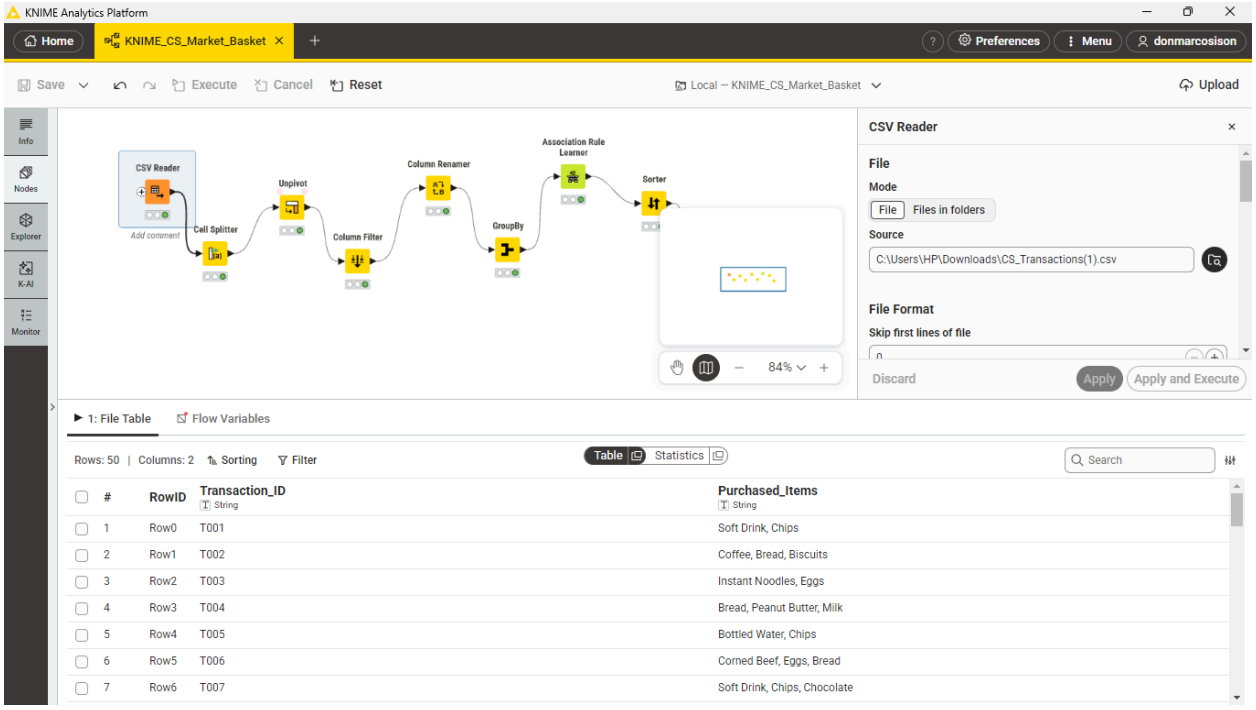
No item should appear twice in the same transaction. 2-5 different products. Transactions 1-50

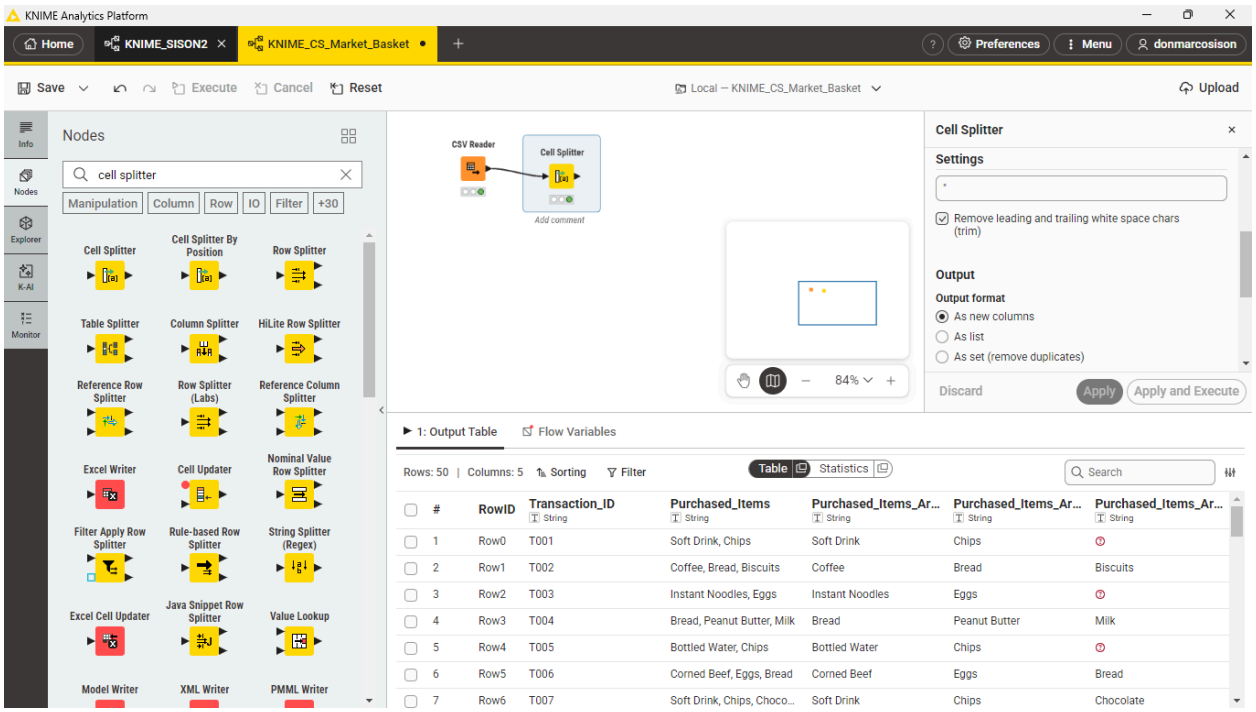
CONVENIENCE STORE TRANSACTIONS	
Transaction ID	**Purchased Items**
T001	Soft Drink, Chips
T002	Coffee, Bread, Biscuits
T003	Instant Noodles, Eggs
T004	Bread, Peanut Butter, Milk
T005	Bottled Water, Chips
T006	Corned Beef, Eggs, Bread

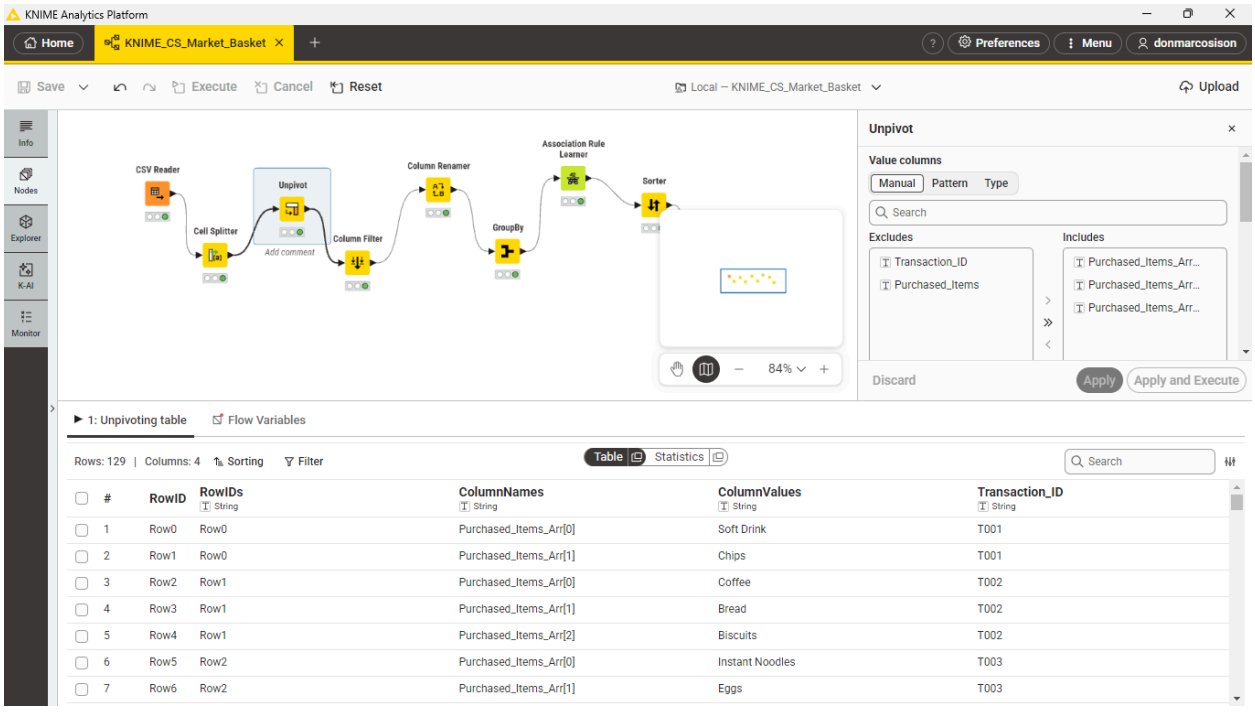
T007	Soft Drink, Chips, Chocolate
T008	Coffee, Bread
T009	Sardines, Eggs, Instant Noodles
T010	Hotdog, Bread, Soft Drink
T011	Coffee, Biscuits
T012	Soft Drink, Chips, Biscuits
T013	Bread, Peanut Butter
T014	Instant Noodles, Eggs, Soft Drink
T015	Bottled Water, Chocolate
T016	Coffee, Bread, Milk
T017	Corned Beef, Eggs
T018	Soft Drink, Chips
T019	Iced Tea, Biscuits, Chocolate
T020	Hotdog, Bread
T021	Instant Noodles, Eggs, Corned Beef
T022	Coffee, Bread, Biscuits
T023	Soft Drink, Chips, Chocolate
T024	Sardines, Eggs
T025	Bread, Peanut Butter, Milk
T026	Bottled Water, Biscuits
T027	Soft Drink, Chips, Bread
T028	Coffee, Biscuits, Chocolate
T029	Instant Noodles, Eggs
T030	Hotdog, Bread, Soft Drink
T031	Iced Tea, Chips
T032	Corned Beef, Eggs, Bread
T033	Coffee, Bread

T034	Soft Drink, Chips, Chocolate
T035	Sardines, Instant Noodles, Eggs
T036	Bread, Peanut Butter
T037	Bottled Water, Chips, Chocolate
T038	Coffee, Bread, Biscuits
T039	Instant Noodles, Eggs, Soft Drink
T040	Hotdog, Bread
T041	Soft Drink, Chips
T042	Coffee, Biscuits
T043	Bread, Peanut Butter, Milk
T044	Corned Beef, Eggs
T045	Iced Tea, Chips, Biscuits
T046	Soft Drink, Chips, Chocolate
T047	Sardines, Eggs, Bread
T048	Coffee, Bread, Milk
T049	Instant Noodles, Eggs
T050	Hotdog, Bread, Soft Drink

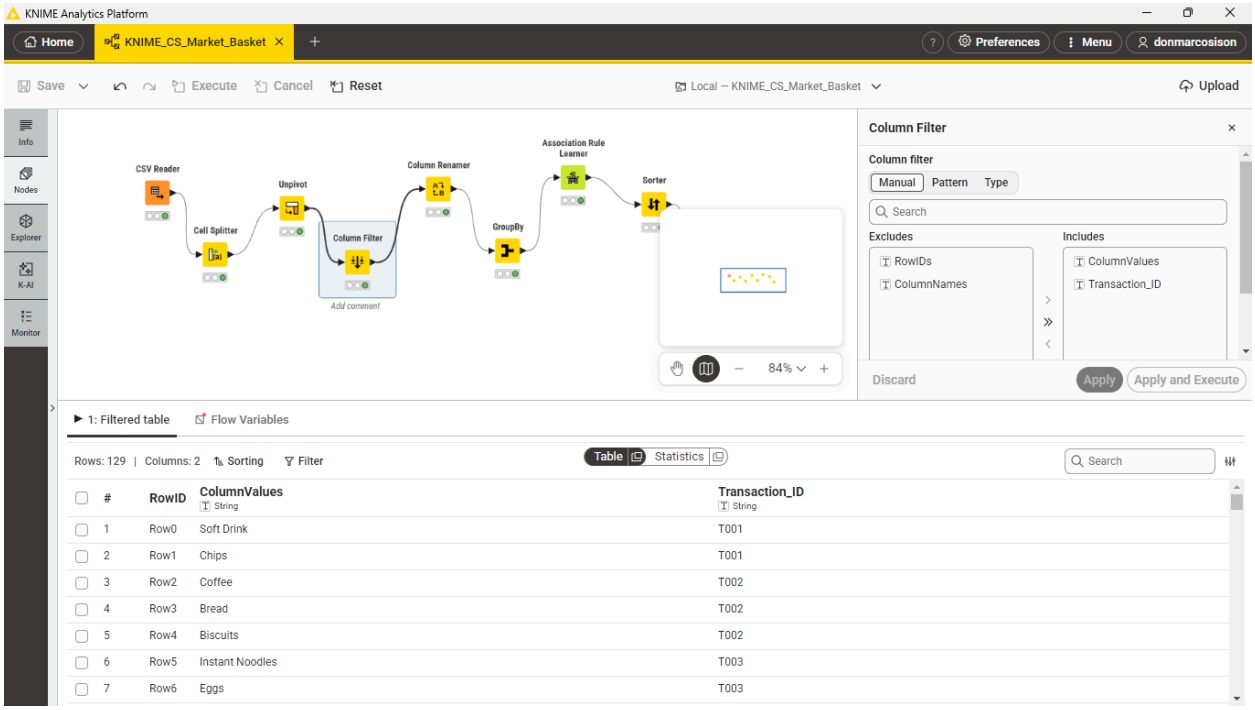
KNIME:


KNIME Analytics Platform
Home
KNIME_CS_Market_Basket
Preferences
Menu
donmarcosison
Save
Execute
Cancel
Reset
Local – KNIME_CS_Market_Basket
Upload
CSV Reader
Unpivot
Column Renamer
Association Rule Learner
Sorter
Cell Splitter
Column Filter
GroupBy
CSV Reader
File
Mode
File
Files in folders
Source
C:\Users\HP\Downloads\CS_Transactions(1).csv
File Format
Skip first lines of file
Discard
Apply
Apply and Execute
1: File Table
Flow Variables
Rows: 50 | Columns: 2
Sorting
Filter
Table
Statistics
Search
RowID Transaction_ID Purchased_Items
1 Row0 T001 Soft Drink, Chips
2 Row1 T002 Coffee, Bread, Biscuits
3 Row2 T003 Instant Noodles, Eggs
4 Row3 T004 Bread, Peanut Butter, Milk
5 Row4 T005 Bottled Water, Chips
6 Row5 T006 Corned Beef, Eggs, Bread
7 Row6 T007 Soft Drink, Chips, Chocolate

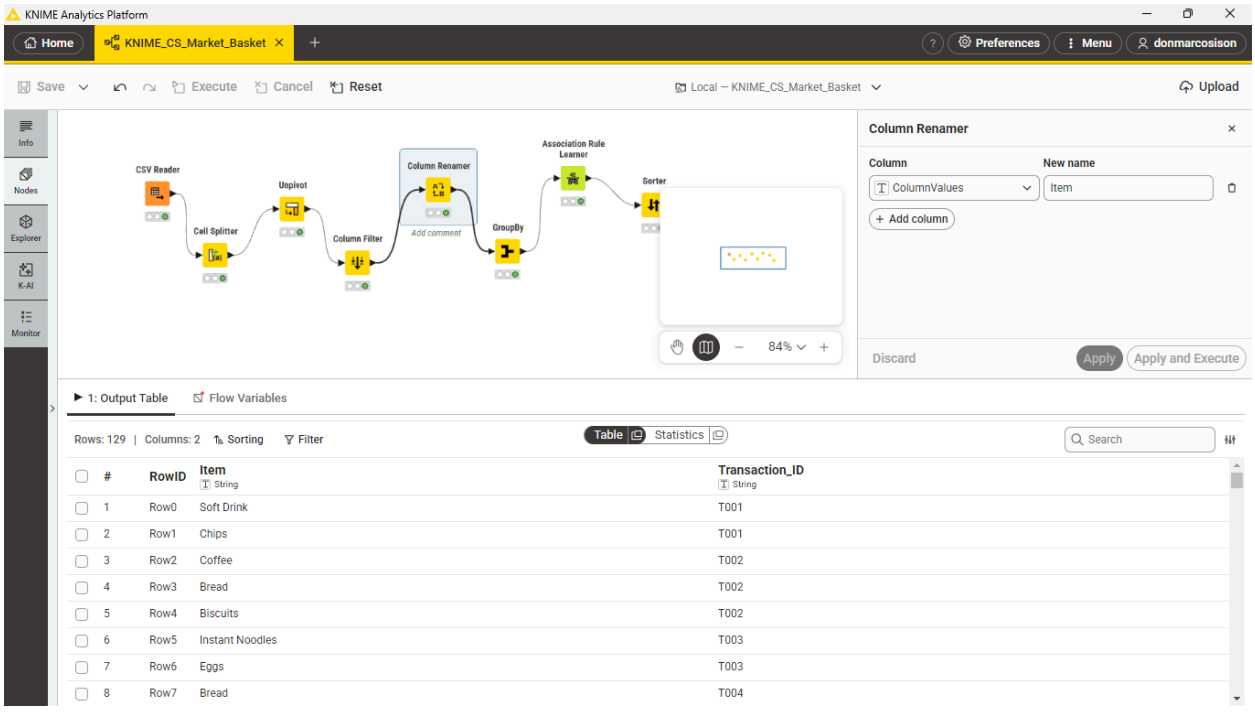

KNIME Analytics Platform
Home
KNIME_SISON2
KNIME_CS_Market_Basket
Preferences
Menu
donmarcosison
Save
Execute
Cancel
Reset
Local – KNIME_CS_Market_Basket
Upload
Nodes
cell splitter
Manipulation
Column
Row
IO
Filter
+30
Cell Splitter
Cell Splitter By Position
Row Splitter
Table Splitter
Column Splitter
HiLite Row Splitter
Reference Row Splitter
Row Splitter (Labs)
Reference Column Splitter
Excel Writer
Cell Updater
Nominal Value Row Splitter
Filter Apply Row Splitter
Rule-based Row Splitter
String Splitter (Regex)
Excel Cell Updater
Java Snippet Row Splitter
Value Lookup
Model Writer
XML Writer
PMML Writer
CSV Reader
Cell Splitter
Add comment
Cell Splitter
Settings
Remove leading and trailing white space chars (trim)
Output
Output format
As new columns
As list
As set (remove duplicates)
Discard
Apply
Apply and Execute
1: Output Table
Flow Variables
Rows: 50 | Columns: 5
Sorting
Filter
Table
Statistics
Search
RowID Transaction_ID Purchased_Items Purchased_Items_Ar... Purchased_Items_Ar... Purchased_Items_Ar...
1 Row0 T001 Soft Drink, Chips Soft Drink Chips
2 Row1 T002 Coffee, Bread, Biscuits Coffee Bread Biscuits
3 Row2 T003 Instant Noodles, Eggs Instant Noodles Eggs
4 Row3 T004 Bread, Peanut Butter, Milk Bread Peanut Butter Milk
5 Row4 T005 Bottled Water, Chips Bottled Water Chips
6 Row5 T006 Corned Beef, Eggs, Bread Corned Beef Eggs Bread
7 Row6 T007 Soft Drink, Chips, Choco... Soft Drink Chips Chocolate

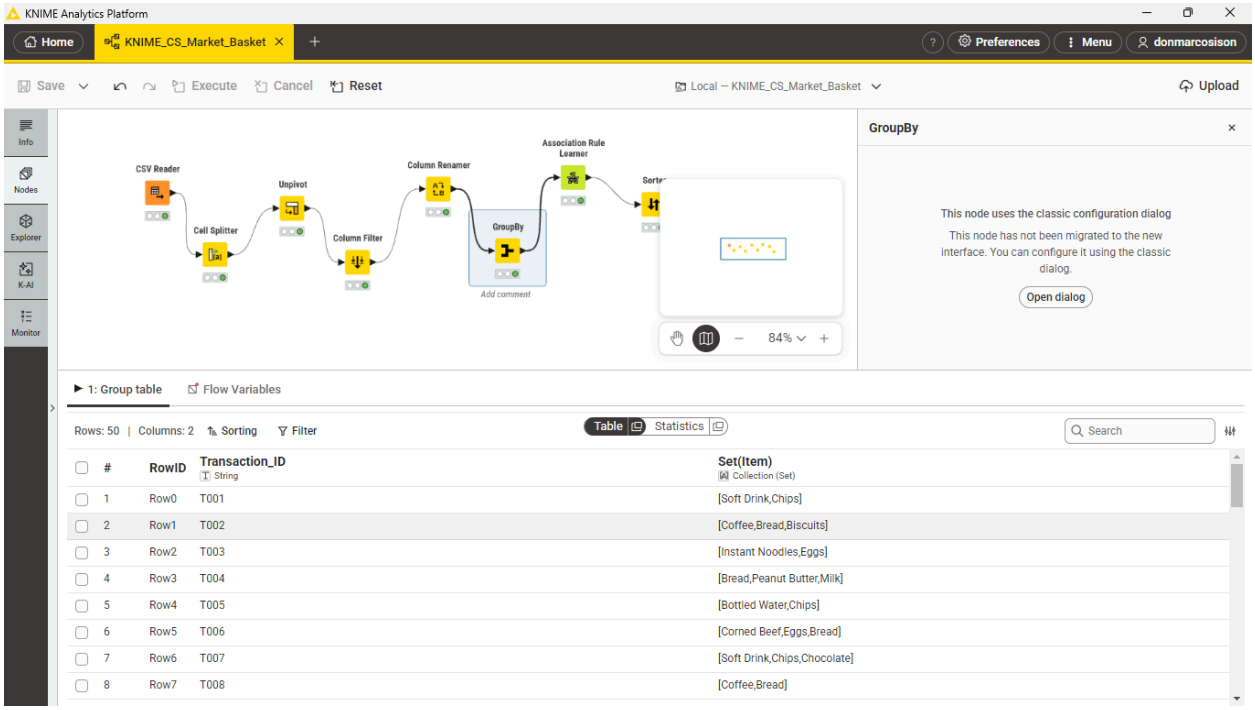


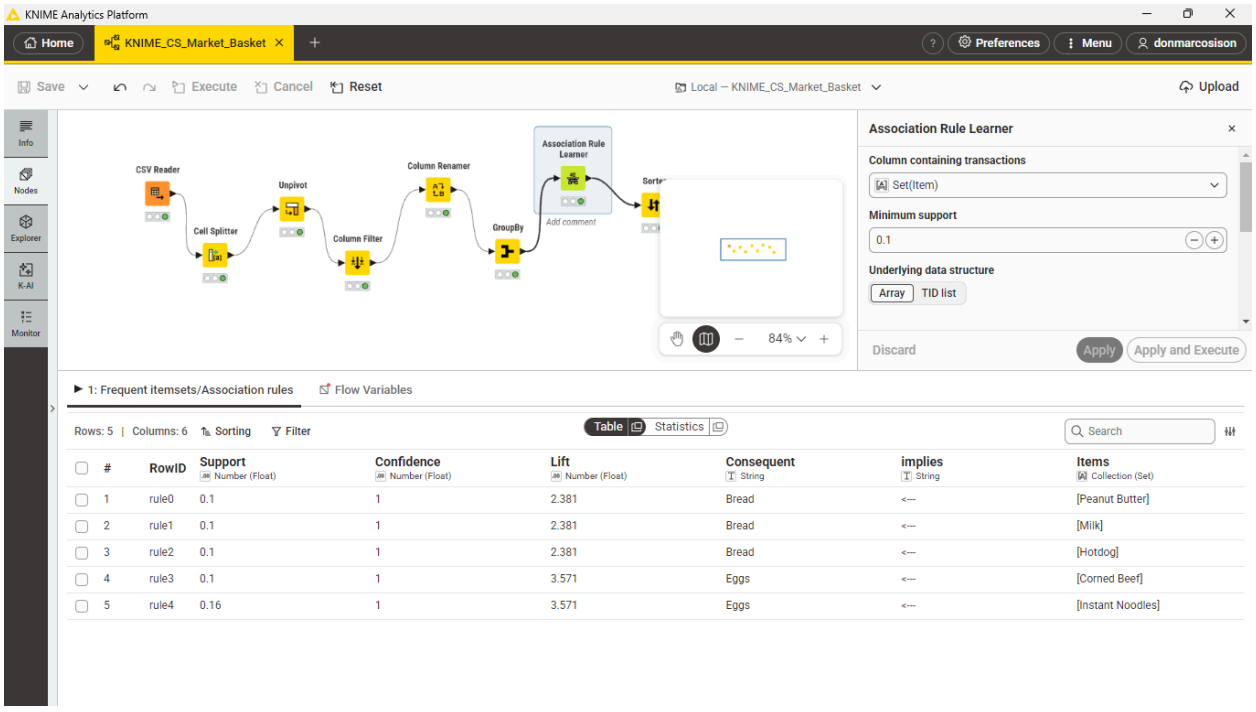
#	RowID	RowIDs (String)	ColumnNames (String)	ColumnValues (String)	Transaction_ID (String)
1	Row0	Row0	Purchased_Items_Arr[0]	Soft Drink	T001
2	Row1	Row0	Purchased_Items_Arr[1]	Chips	T001
3	Row2	Row1	Purchased_Items_Arr[0]	Coffee	T002
4	Row3	Row1	Purchased_Items_Arr[1]	Bread	T002
5	Row4	Row1	Purchased_Items_Arr[2]	Biscuits	T002
6	Row5	Row2	Purchased_Items_Arr[0]	Instant Noodles	T003
7	Row6	Row2	Purchased_Items_Arr[1]	Eggs	T003

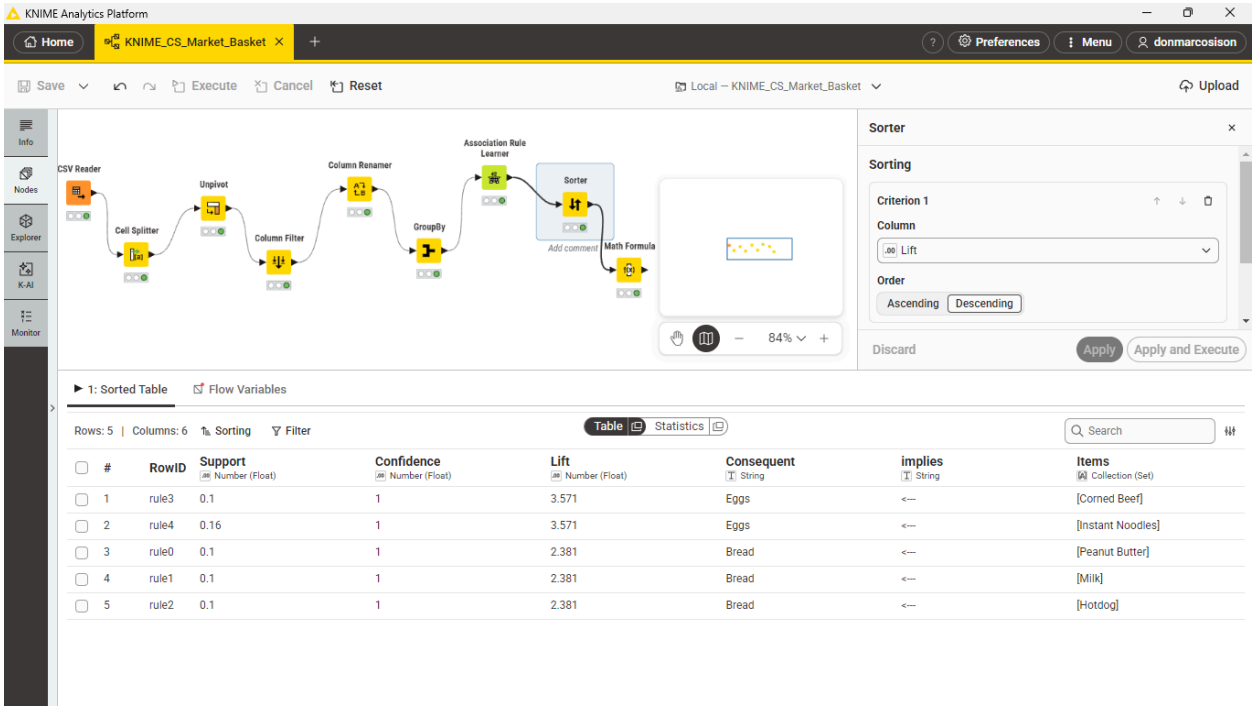


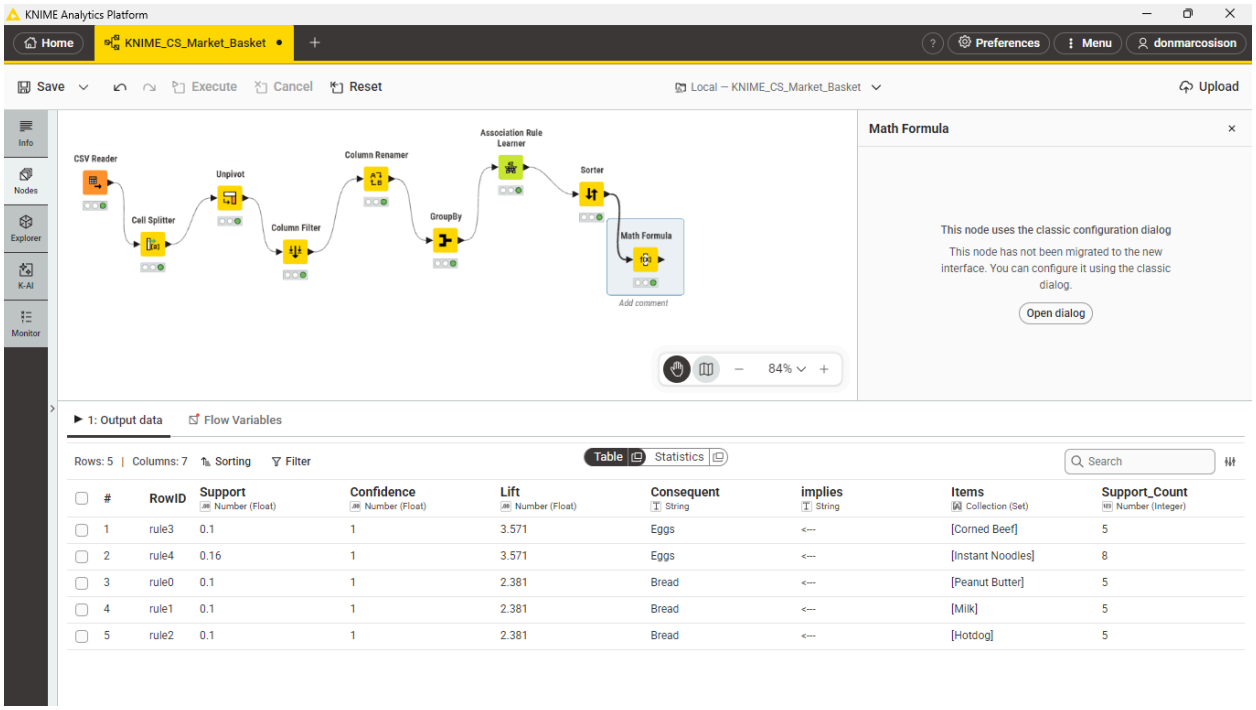
#	RowID	ColumnValues (String)	Transaction_ID (String)
1	Row0	Soft Drink	T001
2	Row1	Chips	T001
3	Row2	Coffee	T002
4	Row3	Bread	T002
5	Row4	Biscuits	T002
6	Row5	Instant Noodles	T003
7	Row6	Eggs	T003

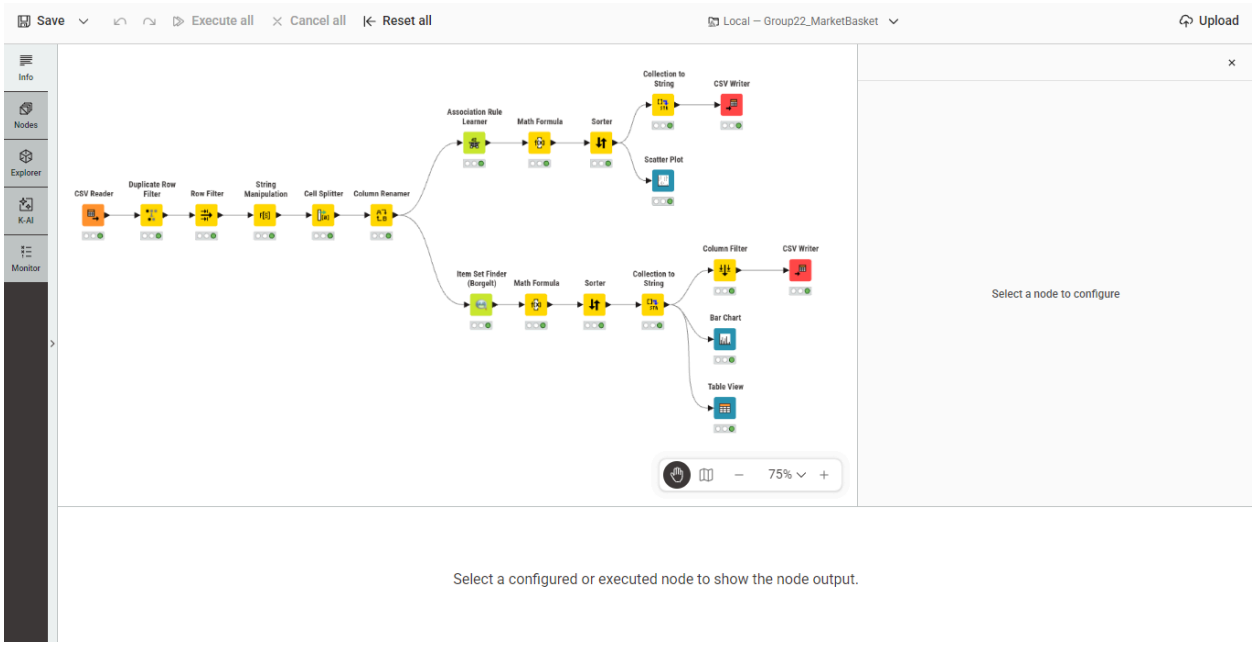

KNIME Analytics Platform
Home
KNIME_CS_Market_Basket
Preferences
Menu
donmarcosison
Save
Execute
Cancel
Reset
Local – KNIME_CS_Market_Basket
Upload
Info
Nodes
Explorer
K-AI
Monitor
CSV Reader
Cell Splitter
Unpivot
Column Filter
Column Renamer
Add comment
GroupBy
Association Rule Learner
Sorter
84%
Column Renamer
Column
New name
ColumnValues
Item
Add column
Discard
Apply
Apply and Execute
1: Output Table
Flow Variables
Rows: 129 | Columns: 2
Sorting
Filter
Table
Statistics
Search
RowID Item String Transaction_ID String
1 Row0 Soft Drink T001
2 Row1 Chips T001
3 Row2 Coffee T002
4 Row3 Bread T002
5 Row4 Biscuits T002
6 Row5 Instant Noodles T003
7 Row6 Eggs T003
8 Row7 Bread T004


KNIME Analytics Platform
Home
KNIME_CS_Market_Basket
Preferences
Menu
donmarcosison
Save
Execute
Cancel
Reset
Local – KNIME_CS_Market_Basket
Upload
Info
Nodes
Explorer
K-AI
Monitor
CSV Reader
Cell Splitter
Unpivot
Column Filter
Column Renamer
GroupBy
Add comment
Association Rule Learner
84%
GroupBy
This node uses the classic configuration dialog
This node has not been migrated to the new interface. You can configure it using the classic dialog.
Open dialog
1: Group table
Flow Variables
Rows: 50 | Columns: 2
Sorting
Filter
Table
Statistics
Search
RowID Transaction_ID String Set(Item) Collection (Set)
1 Row0 T001 [Soft Drink,Chips]
2 Row1 T002 [Coffee,Bread,Biscuits]
3 Row2 T003 [Instant Noodles,Eggs]
4 Row3 T004 [Bread,Peanut Butter,Milk]
5 Row4 T005 [Bottled Water,Chips]
6 Row5 T006 [Corned Beef,Eggs,Bread]
7 Row6 T007 [Soft Drink,Chips,Chocolate]
8 Row7 T008 [Coffee,Bread]

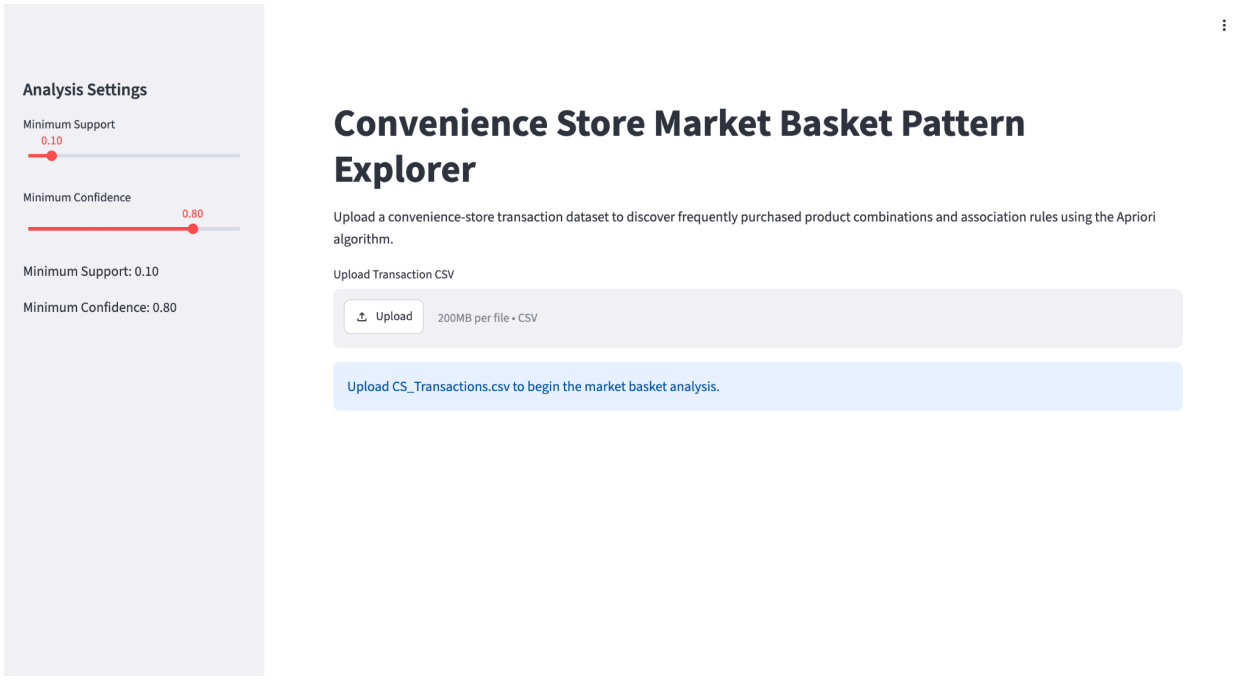

KNIME Analytics Platform
Home
KNIME_CS_Market_Basket
Preferences
Menu
donmarcosison
Save
Execute
Cancel
Reset
Local – KNIME_CS_Market_Basket
Upload
Info
Nodes
Explorer
K-AI
Monitor
CSV Reader
Cell Splitter
Unpivot
Column Filter
Column Renamer
GroupBy
Association Rule Learner
Add comment
Association Rule Learner
Column containing transactions
Set(Item)
Minimum support
0.1
Underlying data structure
Array
TID list
Discard
Apply
Apply and Execute
1: Frequent itemsets/Association rules
Flow Variables
Rows: 5 | Columns: 6
Sorting
Filter
Table
Statistics
Search
RowID Support Number (Float) Confidence Number (Float) Lift Number (Float) Consequent String implies String Items Collection (Set)
1 rule0 0.1 1 2.381 Bread <-- [Peanut Butter]
2 rule1 0.1 1 2.381 Bread <-- [Milk]
3 rule2 0.1 1 2.381 Bread <-- [Hotdog]
4 rule3 0.1 1 3.571 Eggs <-- [Corned Beef]
5 rule4 0.16 1 3.571 Eggs <-- [Instant Noodles]


KNIME Analytics Platform
Home
KNIME_CS_Market_Basket
Preferences
Menu
donmarcosison
Save
Execute
Cancel
Reset
Local – KNIME_CS_Market_Basket
Upload
Info
Nodes
Explorer
K-AI
Monitor
CSV Reader
Cell Splitter
Unpivot
Column Filter
Column Renamer
GroupBy
Association Rule Learner
Sorter
Add comment
Math Formula
Sorter
Sorting
Criterion 1
Column
Lift
Order
Ascending
Descending
Discard
Apply
Apply and Execute
1: Sorted Table
Flow Variables
Rows: 5 | Columns: 6
Sorting
Filter
Table
Statistics
Search
RowID Support Number (Float) Confidence Number (Float) Lift Number (Float) Consequent String implies String Items Collection (Set)
1 rule3 0.1 1 3.571 Eggs <-- [Corned Beef]
2 rule4 0.16 1 3.571 Eggs <-- [Instant Noodles]
3 rule0 0.1 1 2.381 Bread <-- [Peanut Butter]
4 rule1 0.1 1 2.381 Bread <-- [Milk]
5 rule2 0.1 1 2.381 Bread <-- [Hotdog]

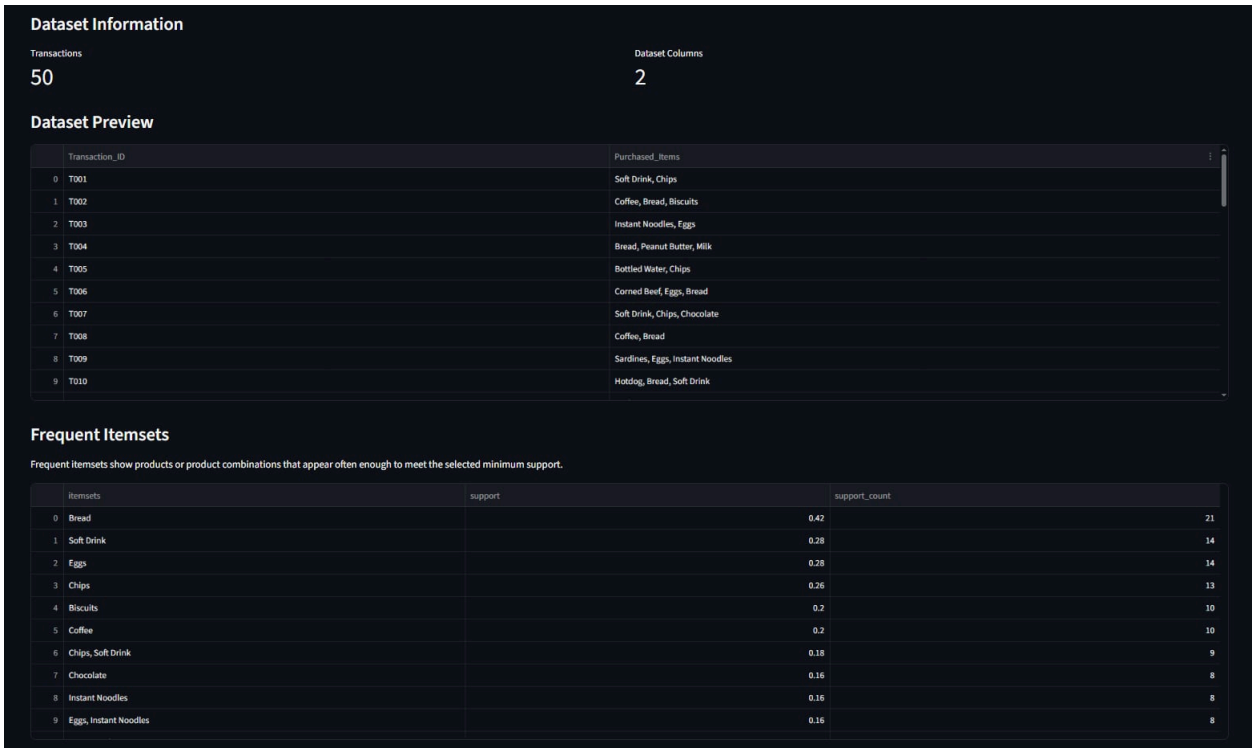

KNIME Analytics Platform
Home
KNIME_CS_Market_Basket
Preferences
Menu
donmarcosison
Save
Execute
Cancel
Reset
Local – KNIME_CS_Market_Basket
Upload
CSV Reader
Cell Splitter
Unpivot
Column Filter
Column Renamer
GroupBy
Association Rule Learner
Sorter
Math Formula
Add comment
Math Formula
This node uses the classic configuration dialog
This node has not been migrated to the new interface. You can configure it using the classic dialog.
Open dialog
1: Output data
Flow Variables
Rows: 5 | Columns: 7
Sorting
Filter
Table
Statistics
Search
| RowID | Support | Confidence | Lift | Consequent | implies | Items | Support_Count
1 | rule3 | 0.1 | 1 | 3.571 | Eggs | <--- | [Corned Beef] | 5
2 | rule4 | 0.16 | 1 | 3.571 | Eggs | <--- | [Instant Noodles] | 8
3 | rule0 | 0.1 | 1 | 2.381 | Bread | <--- | [Peanut Butter] | 5
4 | rule1 | 0.1 | 1 | 2.381 | Bread | <--- | [Milk] | 5
5 | rule2 | 0.1 | 1 | 2.381 | Bread | <--- | [Hotdog] | 5


Save
Execute all
Cancel all
Reset all
Local – Group22_MarketBasket
Upload
CSV Reader
Duplicate Row Filter
Row Filter
String Manipulation
Cell Splitter
Column Renamer
Association Rule Learner
Math Formula
Sorter
Collection to String
CSV Writer
Scatter Plot
Item Set Finder (Borgelt)
Column Filter
Bar Chart
Table View
Select a node to configure
Select a configured or executed node to show the node output.

STREAMLIT:



After uploading CS_Transactions.csv
Result below:



Dataset Information

Transactions: 50

Dataset Columns: 2

Dataset Preview

	Transaction_ID	Purchased_Items
0	T001	Soft Drink, Chips
1	T002	Coffee, Bread, Biscuits
2	T003	Instant Noodles, Eggs
3	T004	Bread, Peanut Butter, Milk
4	T005	Bottled Water, Chips
5	T006	Corned Beef, Eggs, Bread
6	T007	Soft Drink, Chips, Chocolate
7	T008	Coffee, Bread
8	T009	Sardines, Eggs, Instant Noodles
9	T010	Hotdog, Bread, Soft Drink

Frequent Itemsets

Frequent itemsets show products or product combinations that appear often enough to meet the selected minimum support.

	itemsets	support	support_count
0	Bread	0.42	21
1	Soft Drink	0.28	14
2	Eggs	0.28	14
3	Chips	0.26	13
4	Biscuits	0.2	10
5	Coffee	0.2	10
6	Chips, Soft Drink	0.18	9
7	Chocolate	0.16	8
8	Instant Noodles	0.16	8
9	Eggs, Instant Noodles	0.16	8



Association Rules

Association rules show relationships between products using support, confidence, and lift.

	antecedents	consequents	support	confidence	lift	support_count
0	Instant Noodles	Eggs	0.16	1	3.5714	8
1	Corned Beef	Eggs	0.1	1	3.5714	5
2	Hotdog	Bread	0.1	1	2.381	5
3	Peanut Butter	Bread	0.1	1	2.381	5
4	Milk	Bread	0.1	1	2.381	5

This is our workflow below:

Phase 1 — Dataset preparation

Create 50 synthetic transactions in a CSV file.

Assign IDs from T001 to T050.

Check missing values, spelling, and duplicate items.

Create a data dictionary explaining each field.

Field Name	**Data Type**	**Description**	**Example**
Transaction_ID	String	Unique identifier assigned to each convenience-store transaction.	T001
Purchased_Items	String/List	Products purchased together within a single transaction, separated by commas.	Soft Drink, Chips, Chocolate

Phase 2 — KNIME workflow

Import the CSV using CSV Reader.

Convert purchased items into a transaction/item representation.

Generate frequent itemsets using the available association-mining nodes.

Configure minimum support and confidence.

Generate and sort association rules.

Record support counts for the comparison.

Phase 3 — Python implementation

Load and validate the same CSV using pandas.

Convert transactions into one-hot encoded item columns.

Run mlxtend Apriori.

Generate association rules.

Filter rules using support and confidence.

Calculate support count and lift, then export results.

Phase 4 — Streamlit interface

Add CSV upload and a default sample dataset.

Provide minimum support and confidence controls.

Display frequent itemsets and association rules.

Show support, confidence, lift, and support count.

Add downloadable result tables and friendly error messages.

Phase 5 — Testing and comparison

Test normal and missing-input cases.

Test duplicate items and invalid transaction data.

Test thresholds that produce no frequent itemsets or rules.

Compare KNIME and Python results with identical thresholds.

Write a short interpretation and limitations.

Phase 6 — Documentation and presentation

Compile dataset and data dictionary.

Capture KNIME and Streamlit screenshots.

Include the Python implementation and comparison table.

Explain key findings and limitations.

Rehearse each member's explanation.